

VOCALIZATIONS OF BELDING'S GROUND SQUIRRELS (*SPERMOPHILUS BELDINGI*)

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Abstract. Vocalizations of Belding's ground squirrels (*Spermophilus beldingi*) were recorded during the summer of 1982 near Tioga Pass, in the central Sierra Nevada of California. Sonagrams were made and call parameters were measured. Discriminant function analyses revealed that multiple-note calls ('trills') differed acoustically depending on whether they occurred in response to a predator, or were given by males following copulations. The post-copulatory trills of males were individually distinctive. Among anti-predator trills there was no evidence of predator-specificity within the narrow range of predators tested: trills given to two species of (stuffed) weasels (*Mustela*), to dogs and to humans were statistically indistinguishable. Sonagrams of trills occurring in agonistic contexts suggest that a third general category of trills may exist, but agonistic trills were more variable than either anti-predator or post-copulatory trills. The ground squirrels also gave single-note calls in the three contexts described above, either repetitively ('chirps') or singly ('whistles'). Neither chirps nor whistles encoded any obvious situation-specific information, except that whistles were typically associated with rapidly-moving predators, usually raptors. The post-copulatory chirps of males were individually distinctive. Vocalizations of Belding's ground squirrels may not vary among contexts as much as do the analogous calls of California ground squirrels (*S. beecheyi*).

Animal vocalizations, like other forms of communication, are typically variable in structure (Marler 1977; Smith 1977; Byrne 1982; Snowdon 1982). When signal variants are consistently and reliably associated with particular events, activities, individuals or objects, it is reasonable to hypothesize that information about those referents is available in the signals (Andrew 1972; Green & Marler 1979; Hennessy et al. 1981; Smith 1981). For example, vervet monkeys (*Cercopithecus aethiops*) have numerous vocalizations, some of which are given primarily in the presence of predators (Struhsaker 1967). Of these, some are given mainly to snakes, others to large carnivores, and still others to raptors (Seyfarth et al. 1980; Cheney & Seyfarth 1981). Playbacks of calls to free-living vervets support this predator-specification hypothesis, because responses vary with the call type in an apparently adaptive manner (Seyfarth & Cheney 1980; Cheney & Seyfarth 1981). As Marler (1982) and Seyfarth & Cheney (1982) have recently pointed out, the information content of animal vocalizations can suggest how animals perceive or 'categorize' biologically important objects and events.

The information contained in the vocalizations of ground squirrels (Sciuridae: *Spermophilus*) has recently become a subject of considerable interest. For example, California ground squirrels (*S. beecheyi*) apparently give

different alarm calls to (1) snakes, (2) badgers (*Taxidea taxus*), (3) other mammalian predators (including coyotes, *Canis latrans*; bobcats, *Lynx rufus*; and dogs, *Canis familiaris*), and (4) raptors (Leger et al. 1980; Owings & Leger 1980). They also give structurally different calls in other contexts, such as agonistic encounters. Playback experiments conducted in the field have demonstrated that *S. beecheyi* respond to different alarm call types as if the various predators were actually present (Leger & Owings 1978).

Belding's ground squirrels have three main types of vocalizations. Multiple-note 'trills' and repetitive 'chirps' are typically given in response to terrestrial predators, during agonistic encounters, and by males immediately after copulating. Single-note, non-repetitive 'whistles' typically occur in response to raptors or nearby, fast-moving terrestrial predators (Turner 1973; Sherman 1977). These vocalization types are characteristic of many ground-dwelling sciurids (e.g. Betts 1976; Melchior 1971; Koepl et al. 1978; Owings & Virginia 1978; Schwagmeyer & Brown 1981; Nikolski & Wällschlager 1982). Although many aspects of the anti-predator behaviour of Belding's ground squirrels have been studied (e.g. Sherman 1977, 1980a, b), the only work on the acoustic structure of their calls is that of Robinson (1980, 1981). Robinson's study of a population of *S. beldingi* in Oregon suggested that the type of call the animals give may vary

with the eliciting context. In particular, he found that trills given in response to predators differed significantly from those elicited by 'unknown causes' (which Robinson felt were probably not predators). Robinson did not test for the possibility of structural differences among calls evoked by different predatory species.

The purpose of our research was to replicate and extend Robinson's (1980, 1981) results. Specifically, we attempted to determine whether the vocalizations of Belding's ground squirrels vary with the eliciting context. We also investigated the possible predator-specificity of their anti-predator calls, and whether or not there was individual variation in their vocalizations.

Methods

Study Animals and Study Areas

Belding's ground squirrels are diurnal, social rodents that inhabit meadows in the Sierra Nevada and southern Cascade ranges. At high elevations they are active above ground only during the spring and summer months, hibernating the rest of the year. We recorded *S. beldingi* vocalizations at two sites in Mono County, California on 25 different days during June and July 1982. Our high altitude study site (3040 m) is a 1 × 0.5-km subalpine meadow located near Tioga Pass. The area and the ground squirrels' annual cycle of reproductive and social behaviour have been described by Morton et al. (1974), Sherman (1976) and Sherman & Morton (1979). The second study site is similar in size and is located in the lower Lee Vining Creek Canyon, 14 km east of Tioga Pass, at an elevation of 2246 m. The ground squirrels at this site have been studied by Holekamp (1983). Breeding and subsequent reproductive events occur 5–8 weeks earlier in the lower Lee Vining Creek Canyon than at Tioga Pass (Morton 1975; Holekamp 1983).

The behaviour and ecology of the ground squirrels living on Tioga Pass Meadow have been studied for 15 consecutive seasons. As a part of that work, animals are routinely live-trapped and permanently marked with individually numbered metal ear-tags, or by clipping a unique combination of toes. They are also marked with human hair dye on both flanks to facilitate individual identification at a distance. Only a few of the ground squirrels in the lower Lee Vining Creek Canyon population were trapped and dye-marked in 1982.

When our work commenced on 9 June, ground squirrels at Tioga Pass Meadow were just be-

ginning to mate, while those in the lower Lee Vining Creek Canyon had already mated and some of the first litters of the year were emerging from their natal burrows. When data collection ended on 18 July, females at Tioga Pass were still lactating (i.e. pups had not begun to emerge from their natal burrows), while in the lower elevation population the juvenile cohort was active above ground and many adult males and some females had re-entered hibernation.

Equipment and Procedures

Ground squirrel vocalizations were recorded with a Uher 4200 Report Monitor tape recorder (at 19 cm/s) and a Sennheiser ME88 microphone with a K3U power module. For each recorded vocalization we noted the circumstances associated with the call; whenever possible, the caller's identity and sex were also recorded. While recording, we sat on 2–3-m-tall tripods (Tioga Pass), or concealed ourselves behind trees or boulders (both sites).

We used simulated predators to increase the frequency of alarm calling. A stuffed long-tailed weasel (*Mustela frenata*) and a stuffed mink (*M. vison*), both of which are native predators, were each mounted in a quadrupedal position on sleds made of two 2 × 2-cm wooden runners connected by 7-mm wooden dowels front and rear, to which the specimen's feet were attached. Nylon monofilament line was tied to the front dowel so that the sled and predator could be pulled forward from concealment. Hoogland (1981, 1983) used a similar arrangement with a stuffed badger to stimulate alarm calls in black-tailed prairie dogs (*Cynomys ludovicianus*). Calling by Belding's ground squirrels was elicited by hiding the predator in concealing vegetation 1–5 m from a burrow known or thought to contain a study animal. One investigator then waited behind a tree or boulder until the ground squirrel emerged, and pulled the predator into view, while another person operated the recording equipment. The exposure to the simulated predator typically ended a few moments later when the focal animal called and dashed back down the burrow. Twenty-two stuffed mustelid trials were conducted.

We recorded other anti-predator calls by walking dogs (terriers) on 6-m leashes through both study meadows (three episodes), and by recording calls made in response to freely-moving dogs that were not under our control (four episodes). Because calls given in the former situation could

have been elicited by the dog, its handler, or both, we used only calls elicited by freely-roving dogs in our analyses of 'dog-elicited calls'; the 'dog-or-human' calls were used only in comparing anti-predator calls with calls that occurred in other circumstances.

Data Analysis

Sonagrams were made of the recordings on a Voice-Identification 700 Sonagraph using both narrow-band (45-Hz) and wide-band (150-Hz) filters. The narrow-band sonagrams were used in measuring frequency (spectral) variables, and the wide-band sonagrams were used in measuring temporal variables. The original recording was played into the sonagraph at half speed, thereby yielding an effective frequency range of 0–16 kHz. All of the sonagrams were made by one of us (SDB-K), and all sonagram measurements were made by another (DWL). Because information regarding the circumstances and caller's identity were not noted on the sonagrams, all measurements were made 'blind' regarding these factors.

We measured 22 variables from each multiple-note vocalization (Fig. 1A, Plate I): the number of notes; the durations of the first, middle and last notes; the starting and ending frequencies of these three notes; the highest and lowest frequencies of these three notes; the frequency ranges of these three notes (i.e. the difference between the highest and lowest frequencies in each of the notes); and the durations of the first, middle and last inter-note intervals. All measurements were made on the fundamental frequency using a clear plastic ruler. Measurements were rounded to the nearest millimetre, which, after conversion, yielded data that were accurate to within 2 ms for temporal measures and 0.074 kHz for frequency measures.

Because we suspected that some of these 22 variables would be intercorrelated, we used factor analysis with varimax rotation (Nie et al. 1975) on the trill data before proceeding. Varimax rotation simplifies the interpretation of factors because it tends to yield variables that have either high or low (near zero) loadings (i.e. correlations with the factor). We considered a variable to be significantly associated with a factor if the loading was 0.5 or greater. Although this criterion is arbitrary, it falls in the region of commonly-used inclusion criteria (Nunnally 1967, page 292). Our analysis resulted in four factors (Table I). Factor 1 includes variables concerning the frequency characteristics of the middle and last notes; factor 2 includes variables concerning the frequency and duration of the first note; factor 3 includes the inter-note interval variables; and factor 4 includes only middle- and last-note durations and frequency ranges. The number of notes and the frequency range of the first note did not load on any factor by our 0.5 criterion.

Our subsequent multivariate analyses were based on the single variable from each factor that had the greatest loading on that factor. Because variables that load significantly onto a factor are themselves intercorrelated, the choice of one representative from each factor allows us to make statements about the factor without necessitating (redundant) analysis of all the variables. (Although there is a possibility that some information might be lost by using only one variable from each factor, this proved not to be the case. Intercorrelations among same-factor variables were highly significant; furthermore, preliminary analyses revealed no significant increase in discrimination when all the variables were included.) In addition to the four factor-representing

Table I. Summary of Factor Analysis Results on *S. beldingi* Trill Acoustic Variables*

Factor 1	Factor 2	Factor 3	Factor 4
MN start frequency	FN start frequency	FN INI	†MN duration
MN end frequency	FN end frequency	†MN INI	LN duration
MN high frequency	†FN high frequency	LN INI	MN frequency range
MN low frequency	FN low frequency		LN frequency range
†LN start frequency	FN duration		
LN end frequency			
LN high frequency			
LN low frequency			
Non-loaded variables			
†Number of notes			
FN frequency range			

*FN=first note, MN=middle note, LN=last note, INI=inter-note interval.

†Variables retained for use in discriminant function analyses.

variables, we also used the number-of-notes variable, primarily because it is commonly reported in other studies of sciurid vocalizations (e.g. Balph & Balph 1966; Owings & Virginia 1978; Robinson 1981), and also because it is suspected of conveying an 'urgency' message, at least in *S. beecheyi* (Owings & Virginia 1978). Thus our analyses of trill data are based on the following five variables: starting frequency of the last note, highest frequency of the first note, duration of the middle inter-note interval, duration of the middle note, and the number of notes.

The other calls of Belding's ground squirrels consisted of a single pulse of sound emitted either repetitively (chirps) or singly (whistles) (see Figs 1B, 2C); following Leger et al. (1980), we refer to these as 'single-note calls.' When an individual gave a short series of chirps, we analysed the entire set. However, when chirps were given many times by the same individual (sometimes for several minutes at up to 30 chirps per min), we analysed a subset of the notes, taking care to sample from throughout the bout. We measured six variables on the fundamental-frequency sonagram trace of each single-note call: the duration, starting frequency, ending frequency, highest and lowest frequencies, and the frequency range (Fig. 1B).

The five variables from trills and the six variables from single-note calls were analysed with the SPSS discriminant function program using Rao's solution method and *F*-ratios of 4.0 to enter and remove variables (Nie et al. 1975; Hull & Nie 1981). Some of our analyses were performed on a computer-selected random sample of about 40% of all available calls. This was done so that the discriminative function equations could be calculated on a subset of the data, and later tested against the remaining data to assess the accuracy of the classification (Smith et al. 1982). However, when the number of calls was relatively small, we did not use a subsample for analysis because eliminating more than half the data would make discrimination among groups deceptively simple. By retaining all the calls in data-limited situations, the analysis was made more rigorous.

Interpretations of discriminant function analysis results may or may not correspond to perceptual discriminations in the animals that emit and receive the calls. For example, if 20% of a sample of calls is mis-classified statistically, it does not necessarily follow that the animals mis-perceive 20% of their calls. There are several reasons for this. First, statistical analysis is necessarily limited to a small number of the acoustic variables that could be measured from the sona-

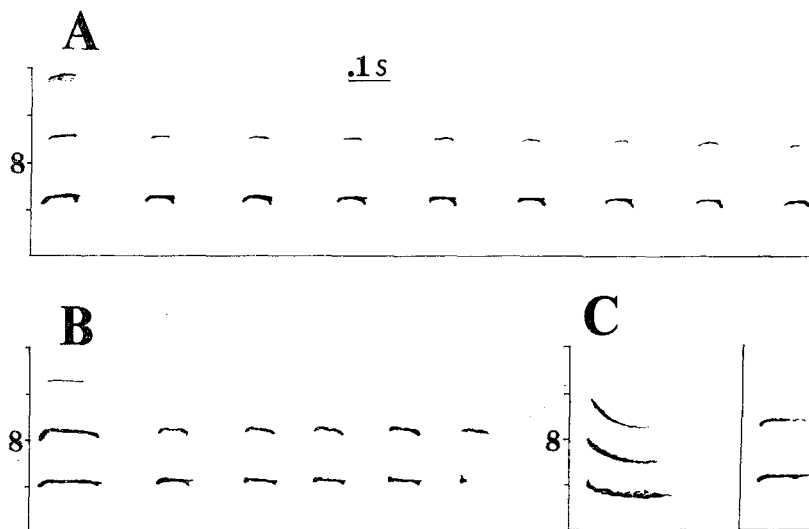


Fig. 2. Narrow-band sonograms of a typical *S. beldingi* anti-predator trill (A) and post-copulatory trill (B), and of anti-predator single-note calls (C: chirp on left, whistle on right). Frequency is scaled in 4-kHz increments.

grams. The animals' auditory systems are not so constrained, however. Second, increasing evidence exists for a perceptual 'dichotomizing' of otherwise continuously graded acoustic variables (Snowdon & Pola 1978; Snowdon 1979, 1982). Thus calls whose acoustic structure falls on or near such a perceptual border may be misclassified by the statistical procedure, but not by the auditory systems of the natural recipients. In short, statistical discriminability may not translate into biological/perceptual discriminability. Only behavioural assays (by means of playback studies, for instance) can confirm or deny our provisional statistical classifications of ground squirrel vocalizations.

Results

Belding's ground squirrels vocalized in a variety of situations, including (1) encounters with aerial and terrestrial predators, (2) agonistic chases and fights with conspecifics, (3) the termination of a copulation (males only), and (4) 'undetermined' causes. In total, we made sonagrams of 312 calls, including 180 trills and 132 single-note calls (Table II). Ninety of the trills and 98 of the single-note calls were recorded at Tioga Pass Meadow. All other calls were recorded in the lower Lee Vining Creek Canyon. The chirps were sampled from approximately 29 separate bouts of calling, and sometimes these bouts had trills interspersed with chirps. We estimate that a minimum of 30–40 individuals contributed the anti-predator calls, 13–15 individuals contributed post-copulatory calls, and about five contributed agonistic calls. Means and standard deviations for the 22 variables we measured for the anti-predator and post-copulatory trills are provided in Table III. Because our sample of agonistic calls is rather small (total $N=26$), we did not attempt to analyse agonistic calls extensively.

Situation-specificity of Calls

Trills. Discriminant function analysis of 42 trills (derived by random selection) revealed significant differences ($P<0.0001$) between anti-predator and post-copulatory vocalizations. The discriminant function equations derived from these 42 calls were used to classify the remaining 77 calls (nine calls were not classified because of missing data for one of the discriminating variables). The result was that 53 of the 77 calls (68.8%) were correctly classified (Table IV). In a second, follow-up classification, we used all the call data (i.e. no sub-sampling) to generate the equations, and then classified all the same calls. In this case 102 of 128 calls (79.7%) were correctly classified (Table IV). This second discriminant analysis included four of the five trill variables (starting frequency of the last note was not used, because it did not account for enough variance to satisfy the inclusion criterion), and these variables were entered in the following order: number of notes, duration of the middle inter-note interval, duration of the middle note, and highest frequency of the first note. Anti-predator trills differed from post-copulatory trills in that the former had more notes, shorter-duration middle notes, longer-duration middle inter-note intervals, and higher-frequency first notes (Table III; all comparisons were significant at $P<0.02$, ANOVA). Anti-predator trills were more variable than post-copulatory trills, perhaps because more individuals were represented in the anti-predator sample.

One possible explanation of our finding that anti-predator and post-copulatory calls are sonographically discriminable is that anti-predator calls were given only by females, whereas post-copulatory calls were given only by males: therefore the situation and the caller's sex are confounded. To resolve this problem we would need to compare male and female calls recorded

Table II. Number of Trill and Single-note *S. beldingi* Vocalizations Recorded from the Major Eliciting Conditions

	Predator	Agonistic	Post-copulatory	Undetermined	Total
Trills	63 (62%)	17 (65%)	65 (45%)	35 (85%)	180
Single-notes					
Chirps	24 (24%)	9 (35%)	77 (54%)	5 (12%)	115
Whistles	14 (14%)	0	2 (1%)	1 (3%)	17
Combined	38	9	79	6	132
Total	101	26	144	41	312

Table III. Means (SD) of Trill Variables for *S. beldingi* Anti-predator and Post-copulatory Trills*

Variable	Anti-predator (N=63)	Post-copulatory (N=65)
†Number of notes	10.5 (7.8)	6.4 (2.6)
FN duration (ms)	85.9 (18.6)	119.1 (19.5)
†MN duration (ms)	61.2 (15.0)	69.2 (12.7)
LN duration (ms)	58.1 (22.5)	70.6 (17.2)
F INI (ms)	151.5 (59.6)	154.0 (37.0)
†M INI (ms)	132.3 (41.8)	119.4 (24.4)
L INI (ms)	136.9 (41.2)	129.7 (37.4)
FN start frequency (kHz)	5.0 (1.1)	4.5 (0.3)
FN end frequency (kHz)	5.1 (1.1)	4.5 (0.4)
†FN high frequency (kHz)	5.6 (1.1)	5.0 (0.5)
FN low frequency (kHz)	4.8 (1.0)	4.2 (0.3)
FN frequency range (kHz)	0.8 (0.2)	0.8 (0.3)
MN start frequency (kHz)	4.7 (0.4)	4.6 (0.4)
MN end frequency (kHz)	4.7 (0.4)	4.5 (0.4)
MN high frequency (kHz)	5.1 (0.4)	4.9 (0.4)
MN low frequency (kHz)	4.4 (0.4)	4.3 (0.3)
MN frequency range (kHz)	0.8 (0.5)	0.7 (0.2)
†LN start frequency (kHz)	4.7 (0.5)	4.7 (0.4)
LN end frequency (kHz)	4.6 (0.6)	4.5 (0.4)
LN high frequency (kHz)	5.0 (0.6)	4.9 (0.5)
LN low frequency (kHz)	4.4 (0.5)	4.3 (0.3)
LN frequency range (kHz)	0.5 (0.2)	0.6 (0.2)

*FN=first note, MN=middle note, LN=last note, INI=inter-note interval.

†Variables used in discriminant function analyses.

Table IV. Classification of *S. beldingi* Trills Based on Discriminant Function Equations

With random selection of cases

Actual group	Predicted group	
	Anti-predator	Post-copulatory
Anti-predator (N=37)	21 (56.8%)	16 (43.2%)
Post-copulatory (N=40)	8 (20.0%)	32 (80.0%)

Without selection (all cases)

Actual group	Predicted group	
	Anti-predator	Post-copulatory
Anti-predator (N=63)	46 (73.0%)	17 (27.0%)
Post-copulatory (N=65)	9 (13.8%)	56 (86.2%)

from the same circumstances. Unfortunately, males rarely call in response to predators (Sherman 1977) and never called during our research, in spite of our exposing them to free-running dogs, a situation that reliably led to calling by females. Similarly, females never called after copulating, so that the relevant male-female comparisons cannot be conducted.

However, from the perspective of an individual that hears a call, the question may be moot: the important issue is whether or not the situation can usually be deduced on the basis of the acoustic properties of the call itself, especially since recipients often cannot see the caller. Moreover, anti-predator (female) and post-copulatory (male) call parameters overlap (Table

III), suggesting that females and males are capable of making sounds with acoustical characteristics that are typical of the other sex.

Taken together, these data indicate that anti-predator and post-copulatory trills, while overlapping to some extent in acoustic structure, are usually discriminable. In fact, after the data were analysed, we usually found it possible a posteriori to assign a call to the eliciting situation by visual inspection of the sonagram (e.g. Fig. 2).

Single-note calls. We found no evidence of situation-specificity in chirps (single-note calls given repetitively). Discriminant function analyses using the six single-note variables (Fig. 1B) revealed no significant differences among chirps given in response to predators or following copulations.

Because most whistles (single-note calls given once) occurred in response to predators (14/17 = 82%), we could not test for situation differences in acoustic structure. However, we did compare 24 predator-evoked chirp notes (derived from five different predator appearances) with 14 predator-evoked whistles (derived from six predator appearances) (Fig. 2C). We discovered significant acoustic differences between the two call types in four of the six variables ($P < 0.001$, t -tests; Table V). This suggests that whistles and chirps may be distinct call categories, not merely the same call emitted at different rates.

Anti-predator Vocalizations

Trills. The trills that we recorded were elicited by the stuffed weasel and mink ($N=15$), freely-roving dogs ($N=15$), humans ($N=27$), a running white-tailed jackrabbit (*Lepus townsendi*, $N=1$), and a California gull (*Larus californicus*, $N=1$); the last species is an occasional predator on ground squirrel pups (Holekamp 1983; personal observations). Among these 59 anti-predator trills, we found no evidence of predator specificity. That is, discriminant function analysis, with and without random selection of cases, revealed no significant differences. Also, one-way

ANOVAs on the five trill variables yielded no significant differences among trills elicited by the various predator species. In addition, we conducted discriminant function analyses on (1) stuffed mustelids versus dogs versus humans, (2) stuffed mustelids versus dogs, and (3) dogs versus humans. In none of these analyses did significant differences appear.

Single-note calls. There was also no evidence of predator-specificity among chirps. Neither discriminant function analyses nor one-way ANOVAs on the six single-note variables revealed any significant differences among chirps evoked by dogs ($N=3$), humans ($N=8$) or gulls ($N=13$); no chirps were recorded in response to the stuffed mustelids.

We also tested for predator-specificity in a small sample of whistles. For this analysis we compared whistles given in response to terrestrial predators ($N=5$, to stuffed weasels and freely-roving dogs) with whistles evoked by low-flying (ca 10 m) avian predators (total $N=5$ whistles, three to California gulls and two to prairie falcons, *Falco mexicanus*). None of the six single note variables differed significantly (all t s < 1.0), suggesting that Belding's ground squirrels, like *S. beecheyi* (Leger et al. 1980), may not differentiate between terrestrial and aerial predators in their whistles. Thus, while whistles seem to indicate the presence of quickly moving predators (Robinson 1980), they do not necessarily specify the type of predator.

Post-copulatory Vocalizations

Trills. At the conclusion of 16 of the 22 above-ground copulations (73%) we observed in 1982, the male gave a series of trills, chirps, or a combination of them. In most of these 16 cases we identified the caller, enabling us to analyse post-copulatory calls for individual distinctiveness. For each of three different males (numbers 1, 5 and 6) we recorded at least 10 post-copulatory trills; each male's calls were recorded following

Table V. Means (SD) of Single-note Variables for *S. beldingi* Anti-predator Whistles and Chirps, and t -test Comparison Results

Variable	Whistles	Chirps	t
Duration (ms)	88.0 (10.4)	167.8 (30.8)	9.3*
Start frequency (kHz)	4.6 (0.4)	4.7 (0.6)	0.6
End frequency (kHz)	4.6 (0.3)	3.8 (0.6)	4.2*
High frequency (kHz)	5.1 (0.3)	5.0 (0.7)	0.4
Low frequency (kHz)	4.4 (0.4)	3.5 (0.5)	5.4*
Frequency range (kHz)	0.7 (0.2)	1.4 (0.4)	6.7*

* $P < 0.001$, $df=36$.

copulations with at least two different females on at least two different days. Discriminant function analysis (without sub-sampling) showed that the post-copulatory trills of these three males were significantly different: 37 of their 46 (80.4%) post-copulatory trills were correctly assigned to the actual caller (Table VI, Fig. 3A).

Although four of the five trill variables differed significantly among the three males' trills (ANOVA, $df=2, 39$; all $P_s < 0.01$), only two variables met the inclusion criteria for the discriminant function equation. First-note highest frequency was the first variable to enter, followed by the number of notes. Interestingly, first-note highest frequency was the last variable to enter in the anti-predator versus post-copulatory call discrimination: thus, statistically at least, the call parameters used to discriminate among individuals are different from those used to discriminate between contexts. Lillehei & Snowdon (1978) documented a similar phenomenon in the 'coo' vocalizations of stump-tailed macaques, *Macaca arctoides*. Furthermore, the single non-significant variable in the individual discrimination (middle-note duration) was different from the non-significant variable in the context discrimination (last-note starting frequency).

Single-note calls. We recorded between nine and 20 chirps given immediately after copulating (total $N=67$) by five different males (numbers 1, 4, 6, 7 and 8; Fig. 3B) on two or three different days with different females. Of the 10 possible pairwise comparisons among these males, nine were significantly different in our discriminant

function analysis (all $P_s < 0.04$). The non-significant comparison and the least significant comparison involved male number 6, for whom we had recorded a sample of only nine chirps; it is possible therefore that the results were affected by the small sample size for this male. Classification analysis based on discriminant function equations (derived without sub-sampling) resulted in 67.2% of the chirps being correctly assigned to the actual caller (Table VI). All six chirp variables were significantly different across these males in univariate analyses (ANOVA, $df=4, 56$; all $P_s < 0.01$).

Agonistic Vocalizations

Our sample of agonistic vocalizations was too small to permit rigorous comparisons with the other categories of calls: therefore, our treatment of them is primarily descriptive.

Trills. The trills that we recorded in agonistic situations ($N=17$) were quite variable, but seemed to fall into three subgroups. The first ($N=7$) was characterized by a large number of short-duration notes (Fig. 4A, Plate II). All of these calls were given by females following territorial disputes with other females, which involved chasing and occasionally fighting. These calls typically occurred immediately after the interactants had physically separated, and they were given by both winners and losers.

The second group of agonistic trills ($N=3$) showed extensive frequency modulation (Fig. 4B). We refer to these calls as 'squeals'. Our examples were all recorded from males, and all

Table VI. Classification of Male *S. beldingi* Post-copulatory Trills and Chirps Based on Discriminant Function Equations

Discriminant Function Equations

Trills		Predicted caller		
Actual caller	Male 1	Male 5	Male 6	
Male 1 ($N=18$)	15 (83.3%)	3 (16.7%)	0 (0.0%)	
Male 5 ($N=18$)	0 (0.0%)	16 (88.9%)	2 (11.1%)	
Male 6 ($N=10$)	1 (10.0%)	3 (30.0%)	6 (60.0%)	

Chirps		Predicted caller				
Actual caller	Male 1	Male 4	Male 6	Male 7	Male 8	
Male 1 ($N=15$)	11 (73.3%)	1 (6.7%)	1 (6.7%)	2 (13.3%)	0 (0.0%)	
Male 4 ($N=11$)	1 (9.1%)	8 (72.7%)	2 (18.2%)	0 (0.0%)	0 (0.0%)	
Male 6 ($N=9$)	1 (11.1%)	1 (11.1%)	2 (22.2%)	5 (55.6%)	0 (0.0%)	
Male 7 ($N=20$)	1 (5.0%)	0 (0.0%)	1 (5.0%)	18 (90.0%)	0 (0.0%)	
Male 8 ($N=12$)	0 (0.0%)	0 (0.0%)	0 (0.0%)	6 (50.0%)	6 (50.0%)	

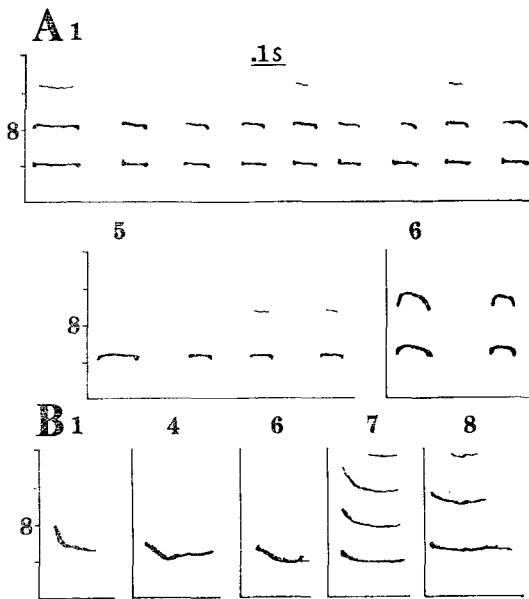


Fig. 3. (A) Narrow-band sonograms of post-copulatory trills by *S. beldingi* males 1, 5 and 6. (B) Post-copulatory chirps by males 1, 4, 6, 7 and 8. The presence of harmonics on some sonograms is probably due to a greater recording gain. Frequency is scaled in 4-kHz increments.

occurred during male-male chases after one male had disrupted another's copulation. On several occasions we also heard females give vocalizations that sounded like squeals, but we were unable to record them.

The third group of agonistic trills ($N=7$) is even less well-defined than the first two. This group consisted of (a) calls by males which closely resembled post-copulatory trills, and (b) calls by lactating females which resembled anti-predator trills (see Fig. 2A, B). Since not all copulations are easily observable, especially if they take place in bushes, burrows, or under rocks, some of these male vocalizations may actually have been post-copulatory calls. Also, because conspecifics are a major cause of pup mortality in *S. beldingi* (Sherman 1981), it is conceivable that females responded to conspecifics that were after their pups with anti-predator calls.

Single-note calls. The small number of agonistic single-note calls ($N=9$) that we recorded were all chirps, although soft whistles were occasionally heard during chases. Agonistic chirps were emitted by both sexes following or preceding chases and fights. Structurally, these

chirps resembled anti-predator (Fig. 2C) and post-copulatory (Fig. 3B) chirps, but our sample was not sufficient to permit detailed acoustic analysis.

Discussion

Intra-specific Comparisons

Prior to our work, the only published analyses of the vocalizations of Belding's ground squirrels were those of Robinson (1980, 1981), based on recordings of 35 trills and 13 chirps from a population in southern Oregon. Some interesting comparisons between our results and Robinson's can be made. First, the Oregon population gave calls with higher frequency notes than did our California population. For example, among anti-predator trills, Oregon *S. beldingi* had mean peak frequencies of 6.4 kHz and 6.5 kHz for the first and second notes respectively, whereas in California the animals had mean peak frequencies of 5.6 kHz and 5.1 kHz for the first and middle notes, respectively. Second, in Oregon the anti-predator trills typically included more notes (mean = 13.1) than in California (mean = 10.5). Third, in Oregon anti-predator trills and trills in 'unknown' situations had first notes whose durations were more than twice as long as the second note. In California the first note was only 40–47% longer than the other notes in anti-predator trills, and 69–72% longer in post-copulatory trills. Finally, the Oregon ground squirrels' trills had notes with a much greater frequency range (about 2 kHz) than did those of the California animals (about 0.8 kHz). Whether these differences are due to sampling error or represent true geographic variation ('dialects'; e.g. Slobodchikoff & Coast 1980) will require additional investigation.

Predator-specificity

Belding's ground squirrels, like many other *Spermophilus* species, typically give multiple-note trill or chirp calls when terrestrial predators approach, and single-note, brief whistles to aerial predators (Turner 1973; Sherman 1977). Although we sometimes observed our study animals responding to terrestrial predators with whistles and to aerial predators with trills, by and large there was a probabilistic association between call type and predator type. However, we found no evidence that the animals made finer distinctions within these broad classes of predators. This is in apparent contrast to California ground squirrels, the only other species in the genus whose vocalizations have been studied in

enough detail to permit comparisons with our data. Owings & Leger (1980; also Leger et al. 1980) found that *S. beecheyi* gave various anti-predator trills that were specific to snakes, badgers, and other predatory mammals.

Alarm calls apparently function in the same context, namely to warn genetic relatives, in both *S. beldingi* and *S. beecheyi* (Sherman 1977, 1980a, b; Leger & Owings 1978; Owings & Leger 1980). Why then do the two species differ in the predator-specificity of their trills? It seems to us that there may be several possible explanations. First, our sample size ($N=63$ trills) may not have been large enough to reveal any predator-specificity that might exist in *S. beldingi* calls. Owings & Leger (1980) inferred predator-specificity from a sample of 109 *S. beecheyi* anti-predator calls. However, 79 of their 109 calls (72.5%) were evoked by dogs; thus, the sample sizes for other predatory species ($N=30$) were similar to ours. It seems unlikely to us therefore that sample-size differences alone can account for the apparent discrepancy.

A second possibility is that the Belding's ground squirrels at our study sites did not react to the stuffed predators as if they were alive. Indeed, our stuffed weasel and mink, although moving, progressed in an unnatural fashion, and obviously did not act or smell like real predators. Our use of models seems unlikely to be the sole explanation for the apparent lack of predator-specificity in *S. beldingi* anti-predator calls, however. First, the ground squirrels almost always called immediately after detecting the model and apparently before studying its movement. To perform a trial, we concealed one of the stuffed weasels in tall grass and when a study animal appeared above ground we pulled the model into view; we never needed to pull the weasel more than about a metre before it was detected, eliciting a call and retreat into the burrow. Second, differences between males and females in the frequency of calling during encounters with free-living predators (Sherman 1977) were also manifested in our trials with the models (see also Hoogland 1983). Nonetheless, only by making detailed comparisons between calls evoked by live weasels and coyotes with those we obtained can the apparent lack of predator-specificity be confirmed. We have not yet accumulated recordings from enough naturally occurring mammalian predator encounters to make such a quantitative comparison.

Third, it seems possible that our *S. beldingi* population is normally exposed to fewer types of

predators than the *S. beecheyi* population observed by Owings & Leger (1980). For example, in 15 years of study, only one garter snake (*Thamnophis*) has ever been seen at Tioga Pass Meadow, and rattlesnakes (*Crotalus*) have never been observed in the lower Lee Vining Creek Canyon. Moreover, the Tioga Pass *S. beldingi* hibernate for 7–8 months per year (Morton & Sherman 1978) and those in the Lee Vining Canyon hibernate for 5–6 months (Holekamp 1983). In contrast, the low elevation *S. beecheyi* population studied by Owings & Leger (1980) is active above ground all year (although there are seasonal activity peaks; Fitch 1948). Thus Owings & Leger's *S. beecheyi* may be faced with a more diverse set of predators, and individuals may encounter predators more often during a year than do our Sierra Nevada *S. beldingi*. If so, one result might be greater selection for specificity in the alarm calls of *S. beecheyi*. To test this hypothesis, it would be worthwhile to compare our results with those from a population of *S. beldingi* which is exposed to a greater diversity of predators, including snakes—e.g. the Oregon population studied by Robinson (1980, 1981). Coss & Owings (in press) have made analogous comparisons among populations of *S. beecheyi* that differ in the kind and number of snake predators, and they found that populations normally exposed to rattlesnakes (*Crotalus viridis*) and gopher snakes (*Pituophis melanoleucus*) treat the two species differently, but populations not normally exposed to rattlesnakes treat the two species similarly.

Finally, another explanation to account for the difference between Owings & Leger's (1980) results and our current data hinges on one aspect of the Owings & Leger data. They reported that three categories of terrestrial-predator trills ('chatters') exist—snake, badger, and other mammalian predators (bobcat, coyote and dog). The badger-evoked calls were few ($N=7$), and were recorded during a single predator appearance. Therefore, they could have been emitted by as few as two individual ground squirrels. Although Owings & Leger took statistical steps to ameliorate these limitations, the possibility remains that the badger-elicited calls were judged to be a distinct category only because they were emitted by idiosyncratic callers. If this is true, Owings & Leger's results condense to only two categories of terrestrial predator trills (snakes versus mammals), and thus would indicate similarities in predator-specificity between *S. beecheyi* and *S. beldingi*.

Individual Differences

Owings & Leger (1980) found individual differences in the alarm calls of *S. beecheyi* to the same predator. Likewise, Koepl et al. (1978) reported considerable inter-individual variation in the alarm chirps of four other *Spermophilus* species. The latter authors suggested that individuality could itself be a message of the calls (sensu Smith 1977). However, they did not specify what, if any, the adaptive significance of individuality in alarm calls might be.

Our alarm trill and chirp data for *S. beldingi* are not sufficient to allow us to investigate individuality in this species. However, we did record enough replicate calls to document individual distinctiveness in post-copulatory trills and chirps (Table VI, Fig. 3). While the functions of such individuality are not known, two hypotheses suggest themselves. First, it seems possible that individuality in post-copulatory calls plays a role in facilitating the re-acquisition of a male's position in the dominance hierarchy. Successful males are usually large and old; they win the majority of their fights with rivals and thus obtain access to receptive females (Sherman 1976). During copulations, which last 8–12 min, males are temporarily absent from mating areas, allowing rivals to infiltrate. By calling after copulating, a male may re-establish his presence and warn conspecifics to depart. If this hypothesis is correct, a male's post-copulatory calls should be especially effective at repelling males that have lost fights to the caller. A second hypothesis is that the calls of males function to indicate mating success to oestrous and pre-oestrous females. Under this hypothesis, females would be attracted to frequent callers because of the advantages of selecting phenotypically (and thus, presumably, genotypically) superior mates as sires for their offspring (e.g. Searcy 1982). Individuality in the calls might enhance the ability of females to discriminate among potential mates. Testing these alternatives offers intriguing possibilities for our future research.

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PLATE I

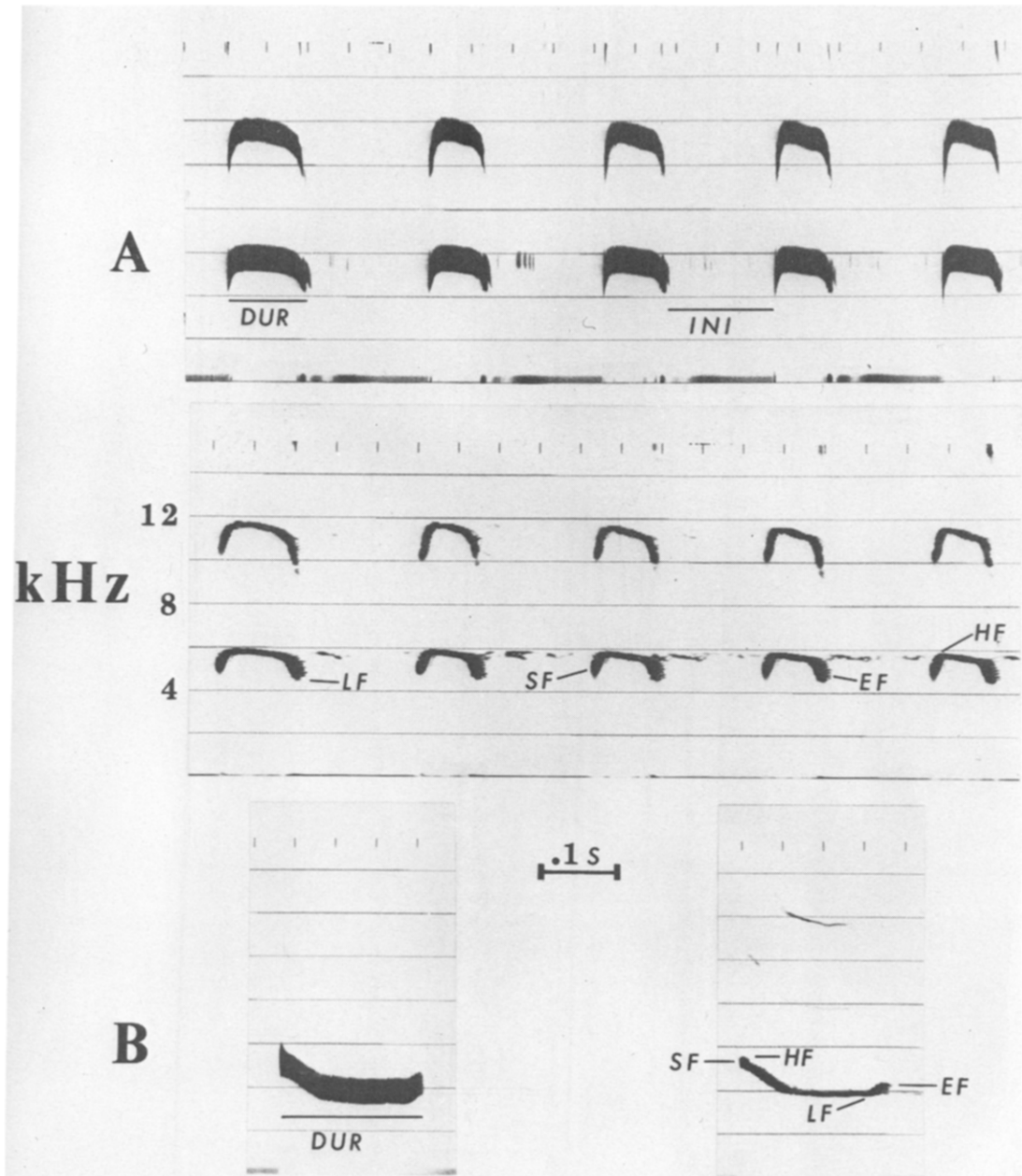


Fig. 1. (A) Wide-band (top) and narrow-band (bottom) sonograms of a five-note *S. beldingi* trill, illustrating some of the variables we measured. DUR=duration, INI=inter-note interval, LF=lowest frequency, SF=starting frequency, EF=end frequency, and HF=highest frequency. (B) Wide-band (left) and narrow-band (right) sonograms of a typical single-note call (a chirp), taken from a series of such notes. Abbreviations as in (A). Horizontal lines through the sonograms are at 2-kHz increments, and the short vertical timing lines at the top of each sonogram are at 50-ms intervals.

PLATE II

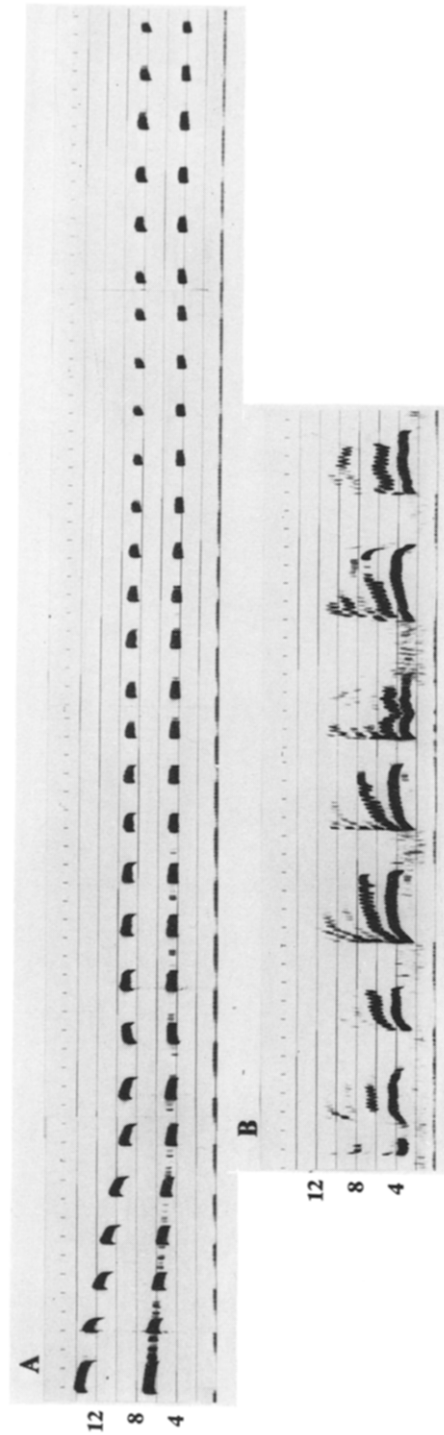


Fig. 4. Wide-band sonograms of an *S. beldingi* female's agonistic trill (A), and a male's agonistic squeal (B). Frequency is scaled in 2-kHz increments; timing marks are at 50-ms intervals.